

Sanghamitra

సంఘమిత్ర

(విజ్ఞాన, వినోద, వికాస ప్రామాణిక పత్రిక)

(A quarterly magazine meant for knowledge, entertainment and progress)

<http://www.sanghamitra.org>

TUESDAY, NOVEMBER 10, 2009



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Volume 3

October 2009

Issue 4

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Foreword

Sanghamitra wishes everyone for Vijaya Dasami (Dussehra) and Diwali. Due to busy schedules of all the volunteers Sanghamitra is being released with few articles this time also. Sanghamitra is looking for some volunteers who would like to proof read the articles and type in the content. We are trying to bring next issue colorfully with variety of articles. Sanghamitra team appreciate your interest and participation. We sincerely thank you all for sending wishes on the occasion of completing 2 years Sanghamitra English version.

Many of our Sanghamitra volunteers attended a meditation program in India, which delayed the last and this issue. We sincerely apologize for the delay. One of our volunteers is contributing for the curriculum of one of the school districts in US. He gave very valuable suggestions to the committee and it is going to be a great achievement if those suggestions are accepted by committee. We post that information in Sanghamitra as soon as we hear about it.

Due to the economic conditions in US and the impact on other parts of the world, many professionals are facing tough times. Hope the economy recovers quickly and this situation of professional (not only IT, but also many manufacturing companies) gets better soon. Hope and confidence are the only things which we should not loose. Let us pray the almighty to give us strength to handle this situation with courage.

Received mails regarding the cover page saying the cover story is good. Many readers suggested us to dedicate the cover page for posting nature and artistic work. We consider their suggestion and try to post such kind of images here onwards. The response to Kids' corner, Salutation to your solution etc. are not up to our expectations. We request readers to encourage kids to answer the various contests.

Thanks for the comments of many scholars regarding the concept of sharing in July issue Foreword. If you want to share your ideas, please send mail to editor@sanghamitra.org or info@sanghamitra.org.

Let us meet again in the next issue due in January.

-Sanghamitra Team.

Seasons of Life!



Seasons of nature.

What is to do with seasons of nature and how do they relate to the seasons of life? What are the seasons of life? These are the questions rising in you. Right?

We are slowly forgetting nature and going away from it. In earlier days people used to relax while enjoying nature. Many of the artists used to fully drown in it and used to come with a nice piece of art work, photos, poetry or some other form. Now we are stuck in frame of daily routine and no creativity most of the time.

It is unfortunate that the animals are also getting trained to forget their nature

and behave like humans in many occasions. Fortunately nature is still trying to continuing its trend. Due to pollution, excavations for petroleum etc. are impacting the nature's cycle also. Even though the environmentalists are warning, we are not caring.

There are Six seasons we used to experience in a year span namely Spring, Summer, Rainy season, Autumn, Fall and Winter. Each one of them corresponds to a stage in our life. Spring is like early childhood. Summer, Rainy season, Autumn, Fall are like Student life, Youth, Married life and later part of life with lots of responsibilities. Winter is the final old age. After Winter again Spring starts. Similarly after death again the soul takes up another body. This is the secret of life.

If we understand this reality, we can enjoy each and every stage of life as well as nature. Hope this brings new understanding in us. This is been shared by our ancestors. We should thank them for their valuable inputs.

Let us meet again in January issue. Till then good bye!

- SuSri.

Words of Wisdom

Perfectly Imperfect

With the blessings of Didiji, publishing her simple and strong words through 'Words of Wisdom', which are capable of transforming a person and lead them towards enlightenment. I know 'Words are powerful', but I experienced the fact in Didiji's presence. Let's say WOW! & Imbibe the 'Words of Wisdom'. May her blessings enlighten many souls and change the world into heaven. Didiji (Shri Siddheshvari Devi) is the founder of Radha Madhav Society. Visit www.radhamadhavsociety.org for a listing of Didi Ji's upcoming lectures and retreats.

The irony of our life situation is that though we are not perfect ourselves, yet we expect others in our life to be perfect. If family members or friends fail us we become very upset. But we must think, "What is the definition of 'perfect' in the first place?" Is a child perfect just because she is getting top marks? Is a perfect wife the one who will smile and say 'yes' to every demand of yours? Is a husband to be considered perfect just because he has the same hobbies as you? Is a perfect friend the one who is always in agreement with everything you say?



Perhaps your son or daughter did not marry the person of your choice. Maybe your wife does not share your hobbies. Your son has perhaps chosen to become a nurse instead of a doctor. It could be that your brother is stuck in a low-paying dead-end job, and your sister is putting up with an abusive husband. Such scenarios are all too common.

If your son marries someone who is not an ideal mate in your eyes, do not cut off all relationship with him and his spouse. It may not be alright in your eyes, but what will you gain by cutting off contact with your own child? How will it help you or your brother if you deride him every time you meet about his lack of ambition? Go ahead and lend a helping hand, but do not judge.

God puts up with all our imperfections. Many of His children denounce Him, do not believe in Him, find fault with Him, but He merely smiles and tolerates. We are not like God, but we can surely learn to accept the imperfections of others, real or perceived.

*

Mathematrixcks

- Sreenivasa Rao Ainapurapu.

In the previous issue we concluded with a question,
find the value of $\left\{ 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \dots + \frac{1}{99} \right\}$

Note : You should teach these techniques to kids only after they are ready to understand these methods. Then only they can appreciate these tips and the concepts behind. It is always better to use basic concrete methods to verify till they are comfortable.

Before answering the question, we need to know little about progressions. What is progression? It is a series of numbers following a formula. Depending on nature we usually come across (i) Arithmetic progression (ii) Geometric progression (iii) Harmonic progression. The given series comes under third category.

What is Arithmetic progression? This follows a pattern where the difference of two consecutive terms is same. For example 1, 4, 7, 10, 13 are in Arithmetic progression. See the common difference $4 - 1 = 7 - 4 = 10 - 7 = 13 - 10 = 3$.

What is Geometric progression? This follows a pattern where the ratio of two consecutive terms is same. For example 1, 3, 9, 27, 81 are in Geometric progression. See the common ratio is $3/1 = 9/3 = 27/9 = 81/27 = 3$.

What is Harmonic progression? This follows a pattern where the reciprocal of each term arranged forms an Arithmetic progression. In our case, the reciprocals of each term forms a series 1, 2, 3, 4, ... 99. We need to figure out the sum of those fractions (which are in Harmonic progression). Harmonic progression does not provide any formula to find out this. So, we need to work on case to case basis.

In this type of situation computer program comes handy. Let us write an algorithm to calculate this. We need to find the reciprocals of numbers from 1 to 99. Let us initiate the SUM to 0. Now we need to follow the steps shown below.

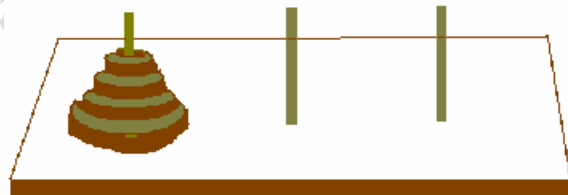
```
SUM = 0;  
For N = 1 to 99  
{  
  // Find the reciprocal value  
  TEMP = 1/N;  
  SUM = SUM + TEMP;  
}
```

Using this algorithm, we can find the value of the specific harmonic progression in our problem. The sum comes to 5.1773775

We can use any programming language like BASIC, PASCAL, C, FORTRAN, C++ etc. While solving problem, we should understand it thoroughly. That is when we write algorithm. Algorithm is sequence of steps which leads to solution.

Let us learn the algorithm for '*Towers of Hanoi*' problem in the next issue.

Towers of Hanoi problem: It is a mathematical game or puzzle. It consists of three rods, and a number of disks of different sizes which can slide onto any rod. Initially the disks stacked in order of size on one rod, the smallest one at the top. The objective of the puzzle is to move the entire stack to another rod, obeying the following rules.



- Only one disk may be moved at a time.
- Each move consists of taking the upper disk from one of the rods and sliding it onto another rod, on top of the other disks that may already be present on that rod.
- No disk may be placed on top of a small disk.

(Let us continue in the next issue, till then good bye!)

Smile if you wish

DO NOT WALK!

A pedestrian was running across the road when there was '**DO NOT WALK**' sign. The other pedestrian was puzzled and asked the first one –

Pedestrian 2 : Hello! Why did you run across the road when the 'Do not Walk' sign is on.

Pedestrian 1 : What was wrong with that?

Pedestrian 2 : You are not supposed to walk when that sign was on.

Pedestrian 1 : That is why I did not walk. I ran quickly!

Pedestrian 2 : ??? ☹



Xerox Copy!

Operator : Sir! May I help you?

Customer: Thank you! Can I have a copy of this document?

Operator : Yes sir! Have a seat sir. (He gets the copy & original in few minutes.)

Customer starts staring at the copy and original document.

Operator : Any problem, Sir?

Customer: Not yet. I am comparing the Xerox to original!

Operator : ?!@# \$

Crazy Boss!

A new energetic, arrogant person was given Manager role. Even though that person is not capable, he started showing off. The workers were little scared to deal with this tough guy. The new manager came out to the lobby and noticed a person carefully observing the notice board.

Boss : Hay! Man, what are looking into?

Person : Trying to figure out the coming holidays list.

Boss : You are supposed to work and finish your job. You are wasting time just looking into notice board. We do not need people who simply spend time during office hours. (He gets wild and says) I am terminating you at this minute , take this amount (pays a month salary) and leave immediately!

Without any response that person leaves.

Boss : (to one of his subordinates) Who is that guy? What was his roll here?

Sub-ordinate : He is a pizza guy. He came to deliver Pizza!

Boss : ? ☹ ?

Future Tense!

Teacher : "I sleep" – change this into future tense.

Student : "You woke up!"

Teacher : ☹

X-Ray!

Patient : Doctor! What does the X-ray of my head show?

Doctor : I'm glad! It doesn't show anything.

Patient : ☹

Let us see some more jokes in next issue. Till then, Good Bye!

Salutation to your solution!

- Sudhakar Attili.

1. There is a rectangular grass field having 15 meters length and 8 meters breadth. If a cow is tied in one corner with a 7 meter long string, how much area can it reach?
2. Can you write $0.\overline{37}$ into fraction? Explain.
3. There is a second degree expression $Ax^2 + Bx + C$. The value of expression is 10 when $x=1$; 5 when $x=0$; 4 when $x = -1$. Find the values of A, B, C and the expression?
4. In the adjacent figure A, Can you move only one coin and make two rows of five coins each?
5. A multi-story building is having floors in such a way that every floor height is $99/100^{\text{th}}$ of its lower level floor height. The ground level floor being 15ft high and the height of the building is 59.105985ft. Find the number of floors in that building?
6. If $(4\sqrt{2})^n = 2^{10}$, what is the value of n?

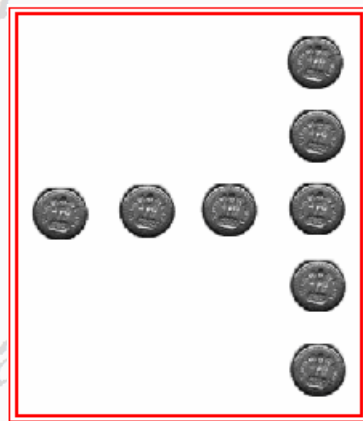


Figure A

You can send your answers either by post or by e-mail to the below address on or before **December 15th**.

Mailing Address:

Editor, Sanghamitra,

e-mail :

Can you figure out?

How do you read 'R U BZ? Y R U late then?'

See the answer with a magnifying glass here →

Third box satisfy the condition "the sum of numbers in each row, column and diagonal should be same"

8	1	6		12	5	10		20	6	16
3	5	7	+	7	9	11	=	10	14	18
4	9	2		8	13	6		12	22	8

Kids' Corner

- Sudhakar Attili.

Unity

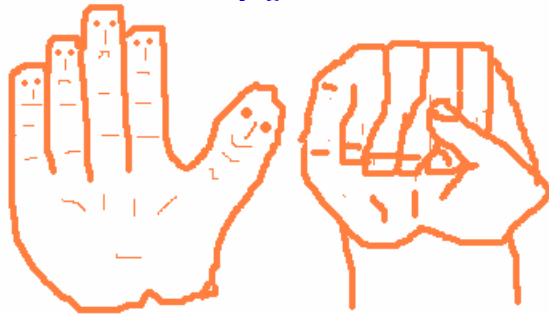
What is Unity? Living together comfortably sharing the resources without any greed. In today's world we see fight between individuals, groups, countries etc. We always fight saying I am great. In reality every one is unique and great in one or other area. We should respect others and should learn how to appreciate others. Let us learn from a small story.

Once there was a fight amongst the five fingers of the hand. Each one claimed that it was the greatest and the most important one.

The little finger said "I am the most important one even though I am the smallest one. When anyone has to do Namaskar I volunteer to stand first and also while counting I am the first."

The ring finger said "I am the greatest because from the king to an ordinary man adore me with a ring studded with precious gem stones. Hence I am called as the Ring finger."

The middle finger said "I am the tallest and I have the honor of having two fingers on either side as body guards."



Like fingers, United we can stand.

Then the fore finger said "I am the ultimate authority while evaluating persons as good and bad."

At last the thumb said with a smile "Oh! What are you all boasting about? You can not do anything without my support."

In fact as we all know each finger is unique in its own way. When the five fingers of the hand come together, then only any task can be accomplished.

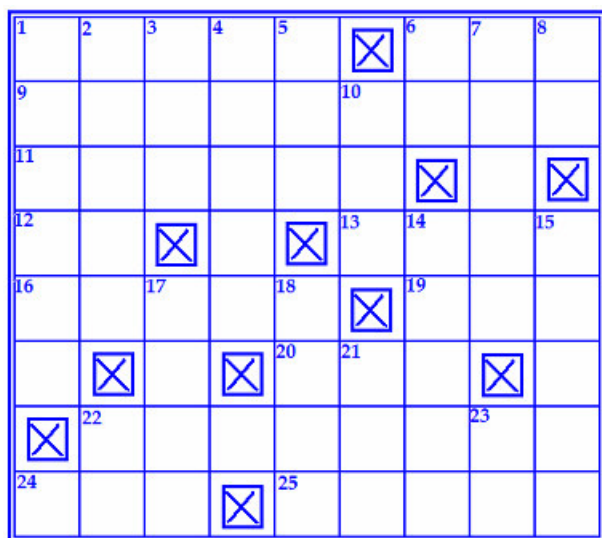
What do we learn from this story?

If every country and religion understand the importance of unity, then there won't be any wars and hatred in the world. Let us practice respecting others' culture and beliefs anticipating the world peace.

Can you find the word?

1. This is a 10 character word and is an item often used to check spelling, meaning, pronunciation etc.
2. 8th, 9th and 4th characters in order form a word which means skilled work or drawing.
3. 7th, 6th, 4th, 8th, 9th and 10th characters in sequence form a word which means authorized seal of a government professional certifying the documents are genuine.
4. 1st, 6th, 7th, 8th, 4th, 5th, 6th and 7th characters in sequence form a word which means 'Charity or generously funding'.
5. 8th, 3rd, 4th, 5th, 6th and 7th characters form a word which means 'pretending, process of doing something'.

Cross Word for kids



Across : 1. Group of eight voices or instruments. (5)

6. Abbreviation of 'Master of Business Administration'. (3)

9. Large Amphibious tropical reptile. (9)

11. Meddle or interfere with. (6)

12. First initials of 'Air Force'. (2)

13. Feeling pity or regret or sympathy.

Duplicate letters are ignored and jumbled. (4)

16. Give or allow as a privilege; Admit to be true. Jumbled! (5)

19. 2nd, 6th and 7th characters of the state FL. (3)

20. First initials of 'One With Almighty'. (3)

22. Wading bird with long legs and pink feathers. (8)

24. The common thing in **bold, sold, fold, told, gold and cold**. (3)

25. For exiting some thing we should first of all should ----. (5)

Down : 1. Hydrocarbon occurring in petrol. (6)

2. Skill, technique. Occupation requiring this. (5)

3. A cartoon character, '--- and Jerry' are rivals. (3)

4. 'War and ----' is a famous novel of Tolstoy. Characters jumbled! (5)

5. While expressing top to bottom in human body we say 'Head to ---'. (3)

6. What is the state abbreviation of Michigan? (2)

7. Heart pumps this throughout the body. (5)

8. Abbreviation of 'Assistant Engineer'. (2)

10. We use this word when something is water free or moist-free. (3)

14. Slope, inclined. (5)

15. Sharp edged instrument used especially for shaving hair from skin. (5)

17. Precious metal ending with '**24 Across**'. (4)

18. Capital of Italy. (4)

21. Be victorious; Gain the favour or support. (3)

22. Two character abbreviation of 'Florida'. (2)

23. Abbreviation of 'General Electric'. (2)

Send your answers (Crossword and 'Can you find the word?') to the below address or e-mail on or before *December 15th*.

e-mail : susri@sanghamitra.org Ph. (314) 395-9516

Mailing Address: Editor, Sanghamitra, 12176 Vivacite Walk, Apt# B, Saint Louis, MO 63146.

Question Gallery

- Sreenivasa Rao Ainapurapu.

1. In which country is the river '**Colorado**' flowing?
2. Who wrote '**Broken wings**'?
3. Where is '**Hoover dam**' located?
4. What is the significance of '**Liver**' in human body?
5. What is the capital of the country '**Germany**'?
6. What is '**Specific heat**'?
7. How many square kilo meters is equal to one '**Square mile**'?
8. What is the symbol for '**Arsenic**'?
9. In which area '**Gopalakrishna Gokhale**' is know for?
10. In which subject do we use the word '**Leguminous**'?

You can send your answers either by postal mail or e-mail to the below address on or before December 15th.

Mailing address :

Editor, Sanghamitra,

or

Miracles

Recently I was sharing my experiences after a meditation program. The seekers were curious to listen what I was going to say. They were looking to hear about miracles happened in that camp. Catching their intension, I asked them "Did you ever experience a miracle?" Only Two hands out of Twenty came up. Everyone became alert to receive the information.

I told them "Everyone of us are experiencing miracles everyday, but we are not realizing them. Without our interaction - our heart, lungs, kidneys are doing a great job. If we get up in the morning with all our senses working, that is a great miracle. Can we deny the fact? What other miracle we are looking for?"

Solutions & Winners

Kids' 'X' Word Solution

1 S	2 T	3 R	4 A	5 N	6 G	7 E	8 R
9 D	10 O	11 N	12 A	13 R	14 O	15	16
17 I	18 P	19	20 G	21 R	22 A	23 S	24 P
25 N	26	27 W	28 E	29 B	30 S	31	32 E
33 V	34 A	35	36 R	37 I	38 S	39 K	40
41 E	42 Y	43 E	44	45	46	47 I	48 L
49 S	50 W	51 O	52 R	53 D	54	55 T	56 O
57 T	58 A	59 F	60	61	62 P	63 E	64 G

Kids Challenge!

Did you find the word?

It is "BALLOON".

Smart kids :

1. Sravya Ainapurapu, St. Louis.
2. Sruthi Ainapurapu, St. Louis.
3. D. Aditya, Lucknow.

Kids' 'X'word Winners:

None. There are few kids who answered the crossword with Two mistakes. Congratulations to them. But, Sanghamitra wishes them a brilliant future. Whether we win or loose, it doesn't count. How much we learnt, that counts.

Salutation to your solution! - Answers

1. Anand beat Vivek by 30 meters, which means - in a given time Anand covered 1500 meters and Vivek covered 1470 meters. Anand beat Swaroop by 100 meters, which means - Anand covered 1500 meters and Swaroop covered 1400 meters. It indicates that in 1470 meters distance Vivek ran 70 meters more than Swaroop. So in the next 30 meters Vivek covers $(30 \times 70) / 1470 = 2100 / 1470 = 10/7$ meters. So Vivek beat Swaroop by $500/7$ meters = 71.42857 meters.

2. TETTFSSSE.

Solution: Pattern is O for One, T for Two, T for Three etc. So the next 9 letters are Ten (T), Eleven (E), Twelve (T), Thirteen (T), Forteen (F), Fifteen (F), Sixteen (S), Seventeen (S), Eighteen (E).

3. 99.

Solution: After 99 cuts, 100 pieces will be available.

4. 25/24.

Solution: $25P = 25 + P$

$$\rightarrow 25P - P = 25$$

$$\rightarrow 24P = 25$$

$$\rightarrow P = 25/24.$$

5. The value of $1/27 = 0.\overline{037}$ and $1/37 = 0.\overline{027}$. The notation with the bar above 0.037 represents the recurring decimal number, which means 0.037037037 and so on.

Solution:

When you multiply 27 and 37, you get 999. In Vedic mathematics the solution is given to this kind of pairs.

6. If $2^n = n^2$, other than 2 there is one integer solution, which is 4.

Salutation to your solution! - Winners:

1. T. Anoop, East Godavari, India.
2. Vaasanti Maghapu, California.
3. Ravi Bhagavatula, Chicago.

Question Gallery! - We ourselves disclosing answers!!

1. Pearl or Guangdong river flows through China.
2. Savitri is Sri. Aurobindo's revelatory mantric epic poem.
3. Canary Wall or Canary Wharf is located in London.
4. Folic acid (also known as Vitamin B9) is especially important during periods of rapid cell division and growth. Both children and adults require folic acid to produce healthy red blood cells and prevent anemia.
5. The capital of Morocco is Rabat.
6. Cryptology deals with the practice and study of hiding information. Modern cryptography intersects the disciplines of mathematics, computer science and engineering. Applications of cryptography include ATM cards, computer passwords and electronic commerce.
7. One Ounce is equal to 28.3495231 grams.
8. Symbol of Selenium is Se.
9. Nikola Tesla (10th July 1856 – 7th January, 1943) was an inventor and a mechanical and electrical engineer. He is best known for his many revolutionary developments in the field of electromagnetism. In the International System of Units unit measuring magnetic field or magnetic flux density or magnetic induction B, which is Tesla, was named in his honor.
10. Phobia is used in Medical terms. Phobia is fear of something, which is totally unreasonable. For example Hypsiphobia is fear of height, Hydrophobia is fear of water, Felinophobia is fear about cats.

Question Gallery! – Winners : 1. Vaasanti Maghapu, California.

2. T. Anoop, East Godavari Dt.
3. Ravi Bhagavatula, Chicago.

Quotes of Swami Vivekananda

- Vedanta does not say 'Give up the enjoyment', but transcend it.
- Do not spend your energy in talking but meditate in silence. Accumulate power in silence and become a dynamo of spirituality.
- If you really want the good of others, the whole Universe may stand against you and can't hurt you. It must crumble before your power of the Lord himself in you, if you are sincere and really unselfish.
- All great undertakings are achieved through mighty obstacles.

Concluding word

Dear readers,

Sanghamitra wishes all readers for Dussehra and Diwali. Sanghamitra is constantly trying to share the knowledge with different groups and in different school districts. Many students are getting benefited by learning our Mathematrix solutions.

As per our ancestors 'Knowledge, Wealth and Joy when shared gets doubled.' We request you to send articles either a science article, news article, about culture or anything that you think is useful for the readers. Send any articles or your feedback to .

We are working on giving prizes to winners by contacting friends' network. Send your answers before December 15th either by E-mail or Postal mail. Encourage your kids to do Kids' Xword and other challenges. Let your friends know about Sanghamitra.

Note : Answers should be sent on or before **December 15th**.

Sanghamitra Team.

Our Address:

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or